Prannaya Gupta

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Education

Georgia Institute of Technology, Atlanta, GA — Aug 2026 – May 2029
Bachelor of Science in Computer Science, Threads: Artificial Intelligence & Embedded Devices

Relevant Coursework: Data Structures & Algorithms, Discrete Math, Linear Algebra, Computer Organization, Operating Systems

National University of Singapore High School of Math and Science, Singapore — Jan 2018 – Dec 2023
Diploma (High Distinction) with Honours in Computer Science and Engineering and Majors in Mathematics, Chemistry and Physics

Work Experience

Pragnition Labs, Singapore – *Research Engineer* — Mar 2026 – Jul 2026
- Prototyped and released medical coding application with React, FastAPI and Pydantic AI within 2 weeks; implemented LLM-powered tree-search inference strategy to categorize between 10,000+ ICD-10-AM and ACHI codes.
- Implemented and evaluated multi-agent debate system for real-time geopolitical analytics and financial evaluations.

Republic of Singapore Air Force, Singapore – *AI Researcher and Team Lead* — Aug 2024 – Feb 2026
- Recruited and grew technical team to 30+ servicemen and interns; led a 15-person subteam developing cognitive monitoring and augmentation systems for air traffic controllers.
- Built and finetuned multimodal representation models using EEG, eye-tracking, and speech signals to estimate cognitive load, stress and fatigue; *research on EEG sensor selection accepted in IEEE EMBC 2026*.
- Led development of *Raider*, a VSCode Extension and multi-agent system for retrieving relevant code snippets from external codebases; trialed during an overseas military exercise. *Follow-up work on modular search published in ICMCIS 2025*.
- Developed domain-adapted speech recognition models for air traffic control communications, *published in ICMCIS 2025*. Implemented Tree-Constrained Pointer Generators for domain adaptation, *initial work accepted in IEEE/AIAA DASC 2026*.

Temus, Singapore – *AI Engineering Intern* — Nov 2023 – Jan 2024
- Built and deployed *TalentForge*, an LLM-powered HR platform using React/Material UI, FastAPI, and GCP; *project awarded the Innovation Award (Industry) at the Google x IMDA Generative AI Trailblazers Competition*.
- Implemented resume summarization and retrieval-augmented chatbot with tool-calling for analytics visualization.

Research Projects

WalledEval: Safety Evaluation Toolkit for LLMs | *HuggingFace, AWS Bedrock, bitsandbytes* — Jun 2024 – Aug 2024
- Developed unified open source evaluation framework for benchmarking frontier and open source LLMs and safety judges across 35+ datasets covering harmful behavior, exaggerated refusals, multilingual safety, and prompt injections.
- Evaluated adversarial robustness under tense, paraphrase, and noise-based prompt mutations, *first-authored resulting work published in EMNLP 2024 System Demonstrations Track*.

Freezing of Gait (FoG) Detection in Parkinson's Patients | *Keras, Scikit-Learn, Android, GCP* — Dec 2020 – Feb 2023
- Developed novel RGB pixelation technique converting triaxial accelerometer time-series into compact images for 2D-CNN-based FoG detection; achieved 98.50% accuracy, outperforming standard model architectures.
- Developed Android application integrating accelerometer sensing with Firebase Cloud Storage to automatically detect and visualize freeze events for patient monitoring. Linked application to Arduino Nano 33 BLE prototype for live monitoring.
- *International Student Science Fair (ISSF) 2023, **1st Place (CS Category)**; Singapore Science and Engineering Fair (SSEF) 2021 and 2023, **Gold Award**; Selected to represent SG at Global Youth Science & Technology Bowl (GYSTB) 2021, **Third Prize**.*

Semantic Segmentation of Interior & Exterior Facades | *PyTorch, ResNet, UPerNet* — Apr 2021 – Feb 2022
- Finetuned UPerNet model for semantic segmentation of 2D apartment interiors and building facades, achieving 49.98% IoU and 82.90% pixel accuracy on 7,400+ images.

Clubs and Societies

Data Science @ Georgia Tech – *Active Project Member* — Aug 2026 – Present
- Optimizing NanoGPT training for a single consumer GPU, experimenting with different model sizes, attention settings, memory- and compute-efficient training strategies to achieve lower validation loss under limited compute and training time.

Skills & Technologies

Programming Languages: Python, JavaScript/TypeScript, Java, Kotlin, MATLAB, GNU Octave, C/C++

AI/ML: PyTorch, Keras, HuggingFace Transformers, Scikit-Learn, bitsandbytes, OpenCV, LangGraph, LiteLLM, DSPy, Pydantic AI

Frameworks & Databases: SQL, PostgreSQL, MySQL, MongoDB, Firebase, Supabase, React, Vue3, NextJS, FastAPI, Tauri